

FGI-GSRx v2.1.0 Release Notes

New Features in v2.1.0

- (1) Support for the FGI-OSNMA Python software package.
- (2) User manual to support integration of FGI-OSNMA with FGI-GSRx. The FGI-OSNMA Python software is called from the MATLAB-based FGI-GSRx software receiver.
- (3) This version will provide “OSNMA strict mode authentication” if used with Galileo E1 signal only:
sys,enabledSignals,[['gale1b']], % List of enabled signals

However, if other signals are enabled, e.g.,

sys,enabledSignals,['gpsl1'] ['gale1b']], % List of enabled signals

This version will provide “OSNMA loose mode authentication” where only Galileo E1 signal would be authenticated, and other signals would not be authenticated.

(4) **Changes: v2.0.2 => v2.1.0**

Folder\Function Name	Remarks	Modification Type
\utils\labsatFileParsing\ls3wToIQ	This function is updated to parse the LabSat datafile with the option to skip some seconds of data from the beginning and also to provide an option to set the duration of the IQ data to be generated.	Modified
\param\ defaultReceiverConfigurationOSNMA.txt	This file provides a default configuration file with the OSNMA parameter settings.	New file added
\osnma\git\ \osnma\app\ \osnma\crypto\ \osnma\datasink\ \osnma\datasource\ \osnma\example\ \osnma\osnma\ \osnma\util\	A new folder called 'osnma' is added to integrate the functionalities of FGI-OSNMA within FGI-GSRx. This folder contains files that are part of the FGI-OSNMA Python software package downloaded from https://github.com/nlsfi/fgi-osnma .	New folder added
\osnma\checkOSNMAObservations.m	This function checks the observations based on OSNMA tag verification. Currently, the function only checks for the ADKD0 authentication flag (status: 1 for successful authentication, 0 for tag failure). It can be easily modified to include other tags.	New function added
\osnma\doOSNMAAnalysis.m	This function reads the text file output of FGI-OSNMA and generates OSNMA tag plots. The identified events are listed as follows: Case 1: Slow MAC self-authentication (ADKD 12) Case 2: Slow MAC cross-authentication (ADKD 12) Case 3: Self authentication (ADKD 0) Case 4: Cross authentication (ADKD 0) Case 5: Galileo timing authentication (ADKD 4) Case 6: ADKD0 authentication failed Case 7: Gap in subframe Case 8: TESLA key verification fails Case 9: All OSNMA bits in a frame are zero Case 10: Tag fail: Not Authenticated Case 11: Authentication not possible: insufficient data with ADKD0 failures Case 12: Missing NAVDATA with multiple ADKD values Case 13: CRC Fail	New function added

	Note: Only features relevant to the receiver are displayed. The user has the option to see multiple other scenarios listed as cases above by uncommenting the relevant section in the plots.	
\osnma\doOSNMAUpdate.m	This function updates the observations with OSNMA verifications.	New function added
\osnma\initiateOSNMA.m	This function is the primary function for initiating the FGI-OSNMA Python software package. It calls the relevant support functions to update the observation data with OSNMA-based information.	New function added
\frame\doFrameDecoding.m	This function has been slightly modified to call OSNMA-related functions.	Modified
\frame\gale1bDecodeEphemeris.m	This function is modified to generate a HEX string that is saved as an ASCII file, which is then used as an input to the FGI-OSNMA block.	Modified
\frame\preambleConsistencyCheck.m	This function performs a consistency check to determine whether all the subframes point to the same subframe beginning.	New function added
\obs\checkObservations.m	This function is modified to call 'checkOSNMAObservations' function.	Modified